Textual Fragmentation and Portfolio Risk

Identifiability and Verifiable Failure Diagnosis

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Revised research manuscript. Recorded experiments, explicit assumptions, and reproducible evidence.

Abstract

Splitting a fixed investment into additional bookkeeping containers should preserve its underlying exposure and risk. Yet apparent invariance can conceal incorrect aggregation: when products have identical holdings, arbitrary nonnegative product multiplicities followed by normalization produce the correct exposure. We characterize this identifiability gap and derive a contrast-basis audit. For four products, three orthogonal composition contrasts identify all free weighting directions of a fixed normalized linear aggregation rule. We give reconstruction, noise and worst-case exposure bounds, while explaining why finite probes cannot certify an unrestricted language model. A cue-free follow-up evaluates the audit against a matched eight-call consistency check on 32 constructed cases in 16 paired families. A separate source-derived tier tests aggregation and source-factored calculation on 20 constructed accounts using five issuer factsheets. All outputs from two model families are retained. For SmolLM3-3B, direct aggregate exposure is correct on 10/32 cases and source-factored extraction on 20/32; the contrast audit has zero coverage. For Qwen3-8B, direct aggregate exposure is correct on 13/32 cases and source-factored extraction on 32/32; the contrast audit has zero coverage. A post hoc extension produces independently checkable numerical certificates that all 32 normalized response traces per model are incompatible with any fixed convex-combination rule within 0.005 error per answer. The contribution is a bounded study of exposure-test informativeness and verification limits, not a new market-risk estimator or evidence of profitable trading.

1 Introduction

An investment account can hold the same underlying assets through one product, several overlapping products, or several funded lots of the same product. Under assumptions that exclude wrapper-specific costs and risks, merely changing that representation leaves the account's payoffs unchanged. This creates a useful test of financial text processing: does a model distinguish the number of described containers from the exposure they collectively represent?

There is a difficulty with the test itself. An answer can remain invariant because the model correctly aggregates holdings, because it emits a constant, or because an incorrect aggregation happens to cancel. In particular, duplicating or omitting identically composed products may be invisible after weights are normalized. More equivalent descriptions do not necessarily provide more evidence about whether every funded lot was processed correctly.

The motivating pilot recorded 472 calls to small Qwen2.5 models. Its numerical results were confounded by a prompt's illustrative unit value, while some apparent improvements from weight normalization were compatible with accidental cancellation. Those observations motivate a fresh follow-up; they are not pooled with the follow-up as confirmatory evidence. The original records and manuscript are retained in the release.

We retain the original research object—portfolio exposure and risk under textual fragmentation—and ask a more precise question within it: *which aggregation errors can an invariance test identify, and what additional queries expose the errors it cannot identify?* We make four bounded contributions:

1. A characterization of the indistinguishability of normalized product-weighting errors on homogeneous holdings, with a rank condition for identifying effective weights from contrasts.
2. A concrete contrast-basis audit, including a closed-form diagnostic, a noise bound, and explicit counterexamples to interpreting finite-query success as a universal certificate.
3. A recorded evaluation with unequal account weights, homogeneous/heterogeneous pairs, matched query budgets, two model families, source-derived holdings, and separate extraction and numerical baselines.
4. A post hoc, exact-checkable falsification of fixed convex-combination explanations, including a recorded global contradiction missed by three specified elementary witness families.

The algebra uses established linear methods. The proposed methodological contribution is their specific use to expose an identifiability problem in portfolio-text evaluation. We do not claim that normalization, finite differences, orthogonal coding, or arithmetic delegation are new inventions.

2 Related work and the contribution boundary

Financial presentation and behavioral testing. Investment-menu effects have long been studied in behavioral finance [1, 6]. They motivate attention to presentation but do not establish that an LLM exhibits the same behavioral mechanism. Buchanan and Foster identify model risk preferences from portfolio choices and report failures of choice invariance [3]; our task instead has a prescribed accounting answer, with no preference to elicit. CheckList provides a general framework for behavioral NLP tests [18]; GSM-Symbolic examines numerical perturbations in mathematical reasoning [15]. Cho et al. systematically study metamorphic relations for NLP [5], and Park's financial metamorphic tests examine filing fidelity in valuation [16]. Accordingly, neither another invariance test nor a presentation-dependent number alone establishes a new financial bias.

Computation and verification. PAL delegates arithmetic to an executable interpreter [8]. Our extraction-plus-calculator baselines follow that established principle. Thokal shows that fixed numerical correction rules can help one financial QA dataset and harm another [19]; this is a direct reason to test normalization outside the cases on which it appears beneficial. VeriFin grounds financial claims in evidence and verifies authorized calculations with a solver [9]. Its evidence-grounding task is broader than the controlled accounting studied here; we do not claim to outperform or replace it.

Perturbation-based diagnosis. ReLoop uses solver responses to parameter perturbations to test LLM-generated optimization models [12]. This is a close methodological precedent: perturbation-based semantic verification is already established. Our narrower contribution is a complete contrast basis for a stated normalized aggregation-error class, together with a demonstration of what homogeneous holdings fail to identify. Encoding numerical computations for error detection also has a long history in algorithm-based fault tolerance [10]. The error model, not the general idea of a checksum, distinguishes the present construction.

Real holdings extraction. TASER extracts structured investment holdings from complex financial tables [4]. Our source tier uses five clean issuer factsheets after table extraction, so it does not reproduce that document-processing difficulty. It tests weighted aggregation of authentic disclosed numbers under constructed account allocations. The appropriate contribution claim is therefore a focused diagnostic study with a source-grounded illustration, not a general solution to financial document understanding.

3 Accounting model and an identifiability gap

Let $q_j \geq 0$ be the account fraction allocated to product j, with $\sum_j q_j = 1$. Let B_{ja} be the fraction of that product in underlying asset a. The aggregate exposure, state return, and standard deviation are

$$w = B^\top q, \qquad R_s = w^\top r_s, \tag{1}$$

$$\sigma(w) = \sqrt{\sum_s \pi_s (R_s - \mu)^2} = \sqrt{w^\top \Sigma w}, \qquad \mu = \sum_s \pi_s R_s. \tag{2}$$

Equation 1 is ordinary weighted accounting; Equation 2 is classical portfolio mathematics [13]. All comparisons fix the joint return distribution and exclude fees, taxes, leverage, trading, tracking error, counterparty risk, and other wrapper-specific effects. These assumptions define a controlled task rather than a claim that actual financial wrappers are economically interchangeable.

Risk accuracy is also weaker than exposure identification. For two assets and $D = R_A - R_B$, classical variance algebra gives

$$\sigma^2(a) = \mathrm{Var}(R_B) + 2a\,\mathrm{Cov}(R_B, D) + a^2\,\mathrm{Var}(D). \tag{3}$$

If $\mathrm{Var}(D) > 0$ and the minimum $a_* = -\mathrm{Cov}(R_B, D)/\mathrm{Var}(D)$ lies inside $[0, 1]$, distinct admissible allocations equidistant from a_* can have exactly the same risk. When $\mathrm{Var}(D) = 0$, all allocations have the same standard deviation. These are standard consequences of the variance formula, not new risk theory. They explain why a correct risk number alone cannot generally certify correct extraction; tolerance-based scoring can obscure further exposure errors.

For two assets, write $b_j = B_{jA}$ and $B_{jB} = 1 - b_j$. The correct account fraction in A is $a = q^\top b$. Consider the restricted error class

$$f_m(b) = \frac{\sum_j m_j q_j b_j}{\sum_j m_j q_j} = p^\top b, \qquad p_j = \frac{m_j q_j}{\sum_k m_k q_k}, \tag{4}$$

where $m_j \geq 0$ and the denominator is positive. Omitting a product corresponds to $m_j = 0$; duplicating it corresponds to $m_j = 2$. More generally m_j represents a relative weighting distortion. We introduce this class as an analytical model, not as an assertion about a language model's hidden computation.

Proposition 1 (Homogeneous-holdings blind spot)**.** *If $b_j = a_0$ for every product, then $f_m(b) = a_0$ for every admissible multiplicity vector m. Neither aggregate exposure nor any risk statistic computed solely from that exposure distinguishes these errors from correct aggregation.*

Proof. Since $\sum_j p_j = 1$, $p^\top(a_0 \mathbf{1}) = a_0$. The aggregate B exposure is $1 - a_0$, so every state payoff and every statistic of those fixed payoffs also coincides. □

The result covers all choices of a_0, not just balanced portfolios. Testing many homogeneous product mixtures at different stock/bond proportions still cannot identify product-specific weighting distortions. Uniform multiplication of all m_j is economically irrelevant after normalization and should not be treated as an exposure error.

For a single heterogeneous composition b, a wrong p may still satisfy $(p - q)^\top b = 0$. Thus correctness on one heterogeneous instance is also insufficient in general. For interior account weights and nonconstant b, normalization and one exact aggregate observation leave, generically, $J - 2$ unconstrained weighting directions. This is an elementary rank observation, not a new result in linear algebra.

4 A contrast-basis exposure audit

4.1 Interventions and conditional identification

Hold account weights, product labels, and sentence structure fixed. For identification, let J count positively funded products; zero-weight products can be removed. For contrast vector c_ℓ, modify each product's A allocation by $\pm\delta c_{\ell j}$ and its B allocation by the opposite amount. Choose δ so every modified allocation remains in $[0, 1]$. These interventions deliberately change economic exposure; they complement the payoff-preserving fragmentation tests rather than pretending to be additional invariant rewrites.

For an observed aggregate-exposure function f, define

$$g_\ell = \frac{f(b + \delta c_\ell) - f(b - \delta c_\ell)}{2\delta}, \qquad s = g - Cq. \tag{5}$$

Under Equation 4 with *the same* p across interventions, $g = Cp$. In particular, the expected response depends on the account allocations q, not the unknown base aggregate answer.

Proposition 2 (Identification from contrast measurements)**.** *Within the class of fixed normalized linear aggregation rules $f(b) = p^\top b$, the contrast measurements $g = Cp$ uniquely identify every admissible weight vector p if and only if*

$$\operatorname{rank} \begin{bmatrix} \mathbf{1}^\top \\ C \end{bmatrix} = J. \tag{6}$$

Consequently, identification from contrasts and normalization requires at least $J - 1$ independent contrast measurements.

Proof. If two normalized vectors produce the same measurements, their difference e satisfies $\mathbf{1}^\top e = 0$ and $Ce = 0$. Full column rank forces $e = 0$. If rank is deficient, take a nonzero null vector e. Around any interior normalized weight vector, a sufficiently small nonzero multiple of e preserves nonnegativity and gives an indistinguishable alternative. The row count gives the lower bound. □

The lower bound is scoped to global identification within this observation scheme. A particular boundary vector can be uniquely identified with fewer measurements because of nonnegativity. The proposition does not assert optimality among arbitrary algorithms supplied with different information or additional ground-truth answers.

For the four-product experiment, use the three nonconstant rows of a Hadamard matrix:

$$C = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}, \quad C^\top C = 4I - \mathbf{1}\mathbf{1}^\top. \tag{7}$$

The effective weighting error and weights have the closed forms

$$\widehat{e} = \tfrac{1}{4} C^\top s, \qquad \widehat{p} = q + \widehat{e}. \tag{8}$$

These identify normalized relative weights, not the absolute scale of multiplicities. Even perfect recovery of p cannot distinguish m from a positive scalar multiple of m.

Corollary 1 (Noise amplification)**.** *If the fixed-linear assumption holds and each queried exposure has additive error of magnitude at most ϵ, then*

$$\|\widehat{p} - p\|_2 \le \frac{\sqrt{3}\epsilon}{2\delta}. \tag{9}$$

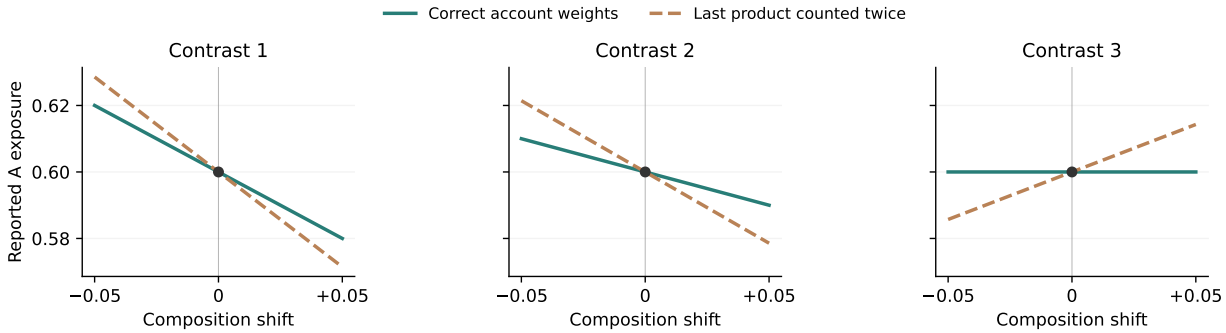


Figure 1: An analytical example with $q = (0.1, 0.2, 0.3, 0.4)$ and homogeneous 60% A holdings. Counting the last product twice gives the same base exposure (black point), but different responses to the three composition contrasts. These lines describe specified linear rules, not observed model outputs.

Proof. Each centered difference has error at most ϵ/δ. The spectral norm of $C^\top/4$ is $1/2$, yielding the stated bound. □

Small perturbations amplify output noise. The experimental choice $\delta = 0.05$ is fixed before the follow-up evaluation; it is not selected to maximize observed success.

4.2 From weighting distortion to potential risk error

An exposure diagnostic is more informative when its magnitude has an economic interpretation. Let $e = p - q$, and restrict within-product A allocations to $[\ell, u]$. Define the worst-case exposure distortion of the fixed aggregation rule over that composition domain by

$$\mathcal{E}(e; \ell, u) = \sup_{b \in [\ell,u]^J} |e^\top b|. \tag{10}$$

Proposition 3 (Sharp exposure bound and a risk consequence)**.** *For normalized p, q,*

$$\mathcal{E}(e; \ell, u) = \frac{u - \ell}{2} \|e\|_1. \tag{11}$$

For two fixed underlying return series R_A, R_B, the corresponding standard-deviation discrepancy satisfies

$$|\sigma(p^\top b) - \sigma(q^\top b)| \le \sigma(R_A - R_B)\, \mathcal{E}(e; \ell, u). \tag{12}$$

Proof. The positive and negative entries of e have equal total magnitude because $\mathbf{1}^\top e = 0$. Assign $b_j = u$ to positive entries and $b_j = \ell$ to negative entries; this attains $(u - \ell)\|e\|_1/2$, and no admissible assignment can exceed it. For the second statement, write the portfolio return as $R_B + a(R_A - R_B)$. The reverse triangle inequality for the norm of centered random variables bounds the difference of standard deviations by $|a_1 - a_2|\sigma(R_A - R_B)$. □

For the four-product basis, exact contrast measurements therefore give

$$\mathcal{E} = \frac{u - \ell}{8} \|C^\top s\|_1. \tag{13}$$

This score distinguishes a small weighting distortion from one capable of materially changing exposure, even if both give the same answer on the homogeneous base case. If the two assets have identical returns, $\sigma(R_A - R_B) = 0$ and exposure mistakes have no payoff consequence in

this restricted model. The risk bound is generally an upper bound, not necessarily attained simultaneously with the sharp exposure bound.

Under Corollary 1, a conservative exposure bound replaces $\|\widehat{e}\|_1$ by $\|\widehat{e}\|_1 + \sqrt{3}\epsilon/\delta$. These are conditional analytical consequences of the fixed-rule model. We do not use them to relabel the empirical audit as a certified market-risk system or retune its frozen acceptance rule.

4.3 Acceptance rule and its limits

The audit uses eight calls: the base description, six centered perturbations, and an anchor with every product entirely allocated to B. For valid fractional outputs, let

$$\kappa = \max_{\ell} \left| \frac{f(b + \delta c_\ell) + f(b - \delta c_\ell)}{2} - f(b) \right|. \tag{14}$$

At the prespecified operating point, the audit accepts the base answer only if

$$\|s\|_\infty \le 0.02, \qquad \kappa \le 0.005, \qquad |f(0)| \le 0.005. \tag{15}$$

Every query must also return valid nonnegative fractions summing to one within 10^{-5}. Missing, malformed, truncated, or invalid outputs cause rejection. The anchor tests an additive offset; curvature tests a local failure of the affine model. The release also reports a tolerance sweep rather than presenting one threshold as universally calibrated.

Even within the fixed-linear class, the empirical slope threshold is not a certificate of the 0.005 accuracy target. For four products, $\|s\|_\infty \le \tau$ implies only $\mathcal{E} \le 3(u - \ell)\tau/4$: the maximum of $\|C^\top s\|_1$ over this cube is 6τ, attained at its vertices. With $[\ell, u] = [0, 1]$ and $\tau = 0.02$, this permits exposure error up to 0.015. The gate is an evaluated operating point, not an analytically calibrated correctness guarantee. Its thresholds remain frozen despite this limitation.

The eight-call implementation is not query-optimal for noiseless linear identification. One-sided contrasts sharing a base query can use fewer calls. We use symmetric pairs to measure curvature and bound difference noise, and a separate anchor to check an additive offset. These additional diagnostics explain the budget; they do not establish universal validity.

Listing 1: Four-product audit. The source structure is supplied by the benchmark.

```
Input: source account weights q, product compositions b
Query: base b; all-B anchor 0;
       b + delta*C[k] and b - delta*C[k], k = 0,1,2
If any answer is invalid or truncated: reject
g[k] = (answer_plus[k] - answer_minus[k]) / (2*delta)
s = g - C*q
curvature = max_k(abs((answer_plus[k] + answer_minus[k])/2
                      - answer_base))
accept = (max(abs(s)) <= 0.02 and curvature <= 0.005
          and abs(answer_anchor) <= 0.005)
diagnostic_weights = q + transpose(C)*s/4
Return: accept, s, curvature, diagnostic_weights
```

This is not a universal semantic certificate. A language model can change its effective computation between prompts. A nonlinear function can agree with the correct function on any finite set of queried points and disagree elsewhere. Curvature and an anchor reduce particular blind spots but do not remove this impossibility. Reconstructed weights are therefore *conditional diagnostics*, not measurements of internal attention or proof that the model omitted a particular product.

The experiment generates interventions from a known source representation. It does not solve the separate problem of reliably editing arbitrary unstructured financial documents. Moreover,

because the experimenter knows q and b, it can calculate the gold answer directly. The audit's role here is to characterize model behavior and benchmark informativeness, not to offer a cheaper replacement for that available calculator.

4.4 An exact-checkable diagnostic of incompatible aggregation

The contrast gate tests whether responses resemble the *correct* weighting rule. Its failure does not establish whether another fixed weighting rule explains them. We therefore add a separately dated, post hoc diagnostic that asks whether *any* nonnegative normalized rule is compatible with the saved answers. This is a bounded-error identification problem in the established set-membership tradition [14]; the optimization uses standard norm approximation and convex duality [2]. The contribution is the portfolio-specific formulation, explicit evidence certificates and empirical diagnosis, not a new linear-programming algorithm.

Let $X \in [0,1]^{T\times J}$ contain the within-product A allocations displayed in the $T = 8$ audit prompts, and let $y \in \mathbb{R}^T$ contain the corresponding reported A exposures. Define

$$d_*(X, y) = \min_{p\geq 0,\, \mathbf{1}^\top p=1} \|y - Xp\|_\infty. \tag{16}$$

This is the smallest uniform answer error needed to reconcile all queries with one admissible aggregation rule. It does not use the prescribed account weights q or the gold aggregate answers. A rule may have $d_* = 0$ and still weight products incorrectly. Conversely, $d_* > \epsilon$ rules out every fixed convex-combination explanation at error budget ϵ; it does not show that every individual answer is wrong or identify a hidden causal mechanism.

Proposition 4 (A verifiable lower bound and an exact dual representation)**.** *For any $\lambda \in \mathbb{R}^T$ with $\|\lambda\|_1 \leq 1$, define*

$$L(\lambda) = \lambda^\top y - \max_j (X^\top \lambda)_j. \tag{17}$$

Then $L(\lambda) \leq d_*(X, y)$, and

$$d_*(X, y) = \max_{\|\lambda\|_1\leq 1} L(\lambda). \tag{18}$$

Any simplex vector $\widetilde{p}$ also supplies the upper bound $d_* \leq U(\widetilde{p}) = \|y - X\widetilde{p}\|_\infty$.

Proof. For any admissible p, the convex-combination property gives $\lambda^\top Xp \leq \max_j (X^\top \lambda)_j$. Therefore $L(\lambda) \leq \lambda^\top (y - Xp) \leq \|y - Xp\|_\infty$ by the dual-norm inequality. Minimizing over p proves the lower bound. For equality, write the infinity norm as a maximum over the unit ℓ_1 ball and interchange the minimum and maximum of the continuous bilinear form on two compact convex sets. Minimizing over the simplex selects the largest coordinate of $X^\top \lambda$. The upper bound follows by evaluating any feasible $\widetilde{p}$. □

The numerical solver proposes primal and dual vectors. We round their coordinates to an integer grid, normalize the primal vector into the simplex, and scale the dual vector to have ℓ_1 norm at most one. A separate checker recomputes $\max(0, L)$ and U using exact rational arithmetic. It requires a bound gap at most 10^{-7}, checks each input against the literal percentages in the archived prompts, and checks each answer against the saved raw response. The checker needs neither an optimizer nor a language model. These are certificates about the printed inputs and saved numerical outputs, not formal verification of the whole software stack.

Simple certificates provide essential comparators. A one-query witness checks whether an answer lies outside the range of its product holdings. A two-query witness checks whether the answer difference lies outside the range of the product-level differences. A three-query midpoint witness checks whether a centered perturbation pair averages to the base response. All are special choices of λ in Proposition 4; the optimizer can combine queries in other ways. The comparison concerns

the stated witness families and error budget, not every possible elementary test. In particular, the original slope check already rejects the cases studied below.

We apply the diagnostic to all evaluation cases in both strict and numerically normalized views. A strict case is eligible only when all eight original fractional outputs are valid. The normalized view reuses the previously defined $a/(a+b)$ transformation; it analyzes transformed answers and does not replace primary scoring. The $\epsilon = 0.005$ reference is the existing exposure tolerance, not an estimated noise level. Because this analysis followed inspection of v4 outcomes, it is exploratory, with no additional model calls or independent sample.

4.5 Comparators and a practical computation baseline

The matched eight-call comparator uses the base description, six lot-order permutations, and a version that splits each of four lots into two. It accepts only if all eight outputs are valid and their A exposures span at most 0.005. This measures representation consistency at the same number of calls. Token and latency costs are also reported; matching calls does not guarantee identical computational cost.

We separately evaluate two one-call computation pipelines. *Aggregate extraction* asks the model for account-level A/B weights, then applies a deterministic risk calculator. *Source-factored extraction* asks for one row per funded lot containing its identifier, account weight, and within-product weights. Software validates identifiers, ranges, account-weight conservation, and within-product conservation before calculating $\sum_j \widehat{q}_j \widehat{b}_j$. These checks detect structural failures but cannot guarantee that values were attached to the correct source clauses. The latter pipeline is an established extraction-and-computation baseline, not the novel part of this work.

5 Experimental design

5.1 Recorded pilot and fresh evaluation

The earlier pilot used Qwen2.5-1.5B and Qwen2.5-3B in Q4_K_M form, 16 initial cases, and an eight-case scale replication. Its 472 recorded calls included post hoc follow-ups. The unit illustration copied by the smaller model and the normalization cancellation are reported as motivating failures, with full details in Appendix D. We do not convert post hoc observations into prespecified findings by rewriting the manuscript.

The follow-up contains 18 newly generated families: two for development and 16 for evaluation. Each family supplies one homogeneous and one heterogeneous four-product description with the same aggregate exposure and return distribution. This yields 32 evaluation cases, not 576 independent samples. Account weights are permutations of $(0.10, 0.15, 0.30, 0.45)$ or $(0.05, 0.20, 0.35, 0.40)$. Base A exposure is sampled between 0.28 and 0.72. Heterogeneous within-product allocations are constructed to preserve that exposure. Three sentence/table grammars vary the order and expression of account versus product weights.

Each case has 18 conditions: base; split lots; six permutations; six perturbations; anchor; source-factored extraction; direct risk generation; and risk generation with verified aggregate weights. Four synthetic joint return states are supplied for risk calculations. They define a descriptive numerical task, not an estimated future market distribution. Risk-changing perturbations are scored on exposure response; risk-generation conditions use the original states and base exposure.

5.2 Models, decoding, and chronology

The fresh evaluation uses Qwen3-8B and SmolLM3-3B, both pinned Q4_K_M artifacts [11, 17]. The release records repository revisions, SHA-256 hashes, runtime command lines, complete API

responses, token usage, and request timestamps. Each condition contributes one response to the main evaluation, at temperature zero, with reasoning disabled, JSON-object constrained decoding, and a 512-token output budget. There is no paid API or best-of-n selection. Original observations are never overwritten; the explicitly defined execution recheck below is stored separately. Infrastructure interruptions required resuming requests without a saved response. Reported denominators count saved records rather than every interrupted attempt.

CPU inference uses llama.cpp b12110. Memory pressure required changes to Qwen3 batching, weight repacking and the server's prompt RAM cache. Table 15 reports every actual profile and its saved-response count. These decisions used resource behavior, not answer accuracy; model bytes, prompts, decoding settings and scoring stayed fixed. Recorded segments identify each response's configuration. Batching and kernels can affect numerical reproducibility, motivating a uniform-configuration recheck.

The stimulus and metric protocol was locally hashed before fresh evaluation responses. Analysis implementation was completed while inference ran, following the recorded plan. This is a follow-up designed with knowledge of the earlier pilot, not an externally preregistered study. Four development requests per new model check execution and formatting; they are retained separately and excluded from evaluation accuracy. The larger output budget and removal of illustrative answer values address two specific pilot limitations, while changes in model, prompt, and decoder prevent attributing any cross-version difference to a single factor.

5.3 Uniform-configuration execution amendment

After 360 original Qwen3 core responses had been saved, we froze a separate execution-recheck protocol selecting all 167 observations whose commands differed from the final runtime profile. Selection used request identifiers and commands only, not correctness. Every selected prompt is re-executed once with eight slots, 1,536 context tokens per slot, weight repacking disabled, and the server prompt RAM cache disabled. The full request payload, model bytes and all scoring rules are unchanged.

The main Qwen3 analysis uses these 167 new responses and the 409 original core responses already generated under that profile. Replacement is mandatory for every selected identifier, even if the new response is worse or invalid. The original mixed-profile analysis and paired differences are retained in Appendix G. Source and challenge responses use the final profile and are not repeated. The final ten challenge calls and the recheck resumed after an interruption and workspace reset, with model and engine hashes verified. Uniformity refers to the command profile, not an independently controlled comparison of host hardware or request batches. This is a dated execution amendment, not an independent sample or an externally preregistered replication. It adds 167 records without increasing the 32-case denominator. Across both models, the release contains 1,400 original evaluation records, 167 execution-recheck records and eight separate development records, in addition to the earlier pilot.

5.4 Source-derived holdings tier

Five official Vanguard LifeStrategy factsheets, reporting holdings as of 30 September 2026, provide 48 disclosed fund-level holding rows [20]. Two PDF text extraction engines agree on all 48 name/percentage pairs, and rendered pages are visually checked within the project. No independent human annotation is claimed. Source URLs, page numbers, dates, file hashes, and the transcribed numerical tables are supplied.

For each of the ten fund pairs, constructed account weights of $(0.30, 0.70)$ and $(0.65, 0.35)$ yield 20 scenarios. The task asks for the account fractions invested in three named underlying funds: developed-world ex-UK equity, UK all-share equity, and emerging-market equity. These are a

subset of account exposures and must not be normalized to sum to one. We compare direct aggregation, split-lot aggregation, and source-factored extraction, giving 60 calls per model.

The published percentages are rounded to one decimal place. We compute gold answers from those displayed values without renormalization. If each published holding is rounded to the nearest 0.1 percentage point, its half-width is 0.0005 in fractional units; a positive weighted average with account weights summing to one has at most the same half-width. The all-target accuracy tolerance is 0.0006. This is a display-precision calculation, not an estimate of market uncertainty. The input is a source-derived holdings table, not an original narrative document or actual client's account. Cases share documents, so we do not treat 20 scenarios as 20 independent financial sources.

5.5 Prospectively frozen challenge extension

During core inference, we added a separate challenge protocol to distinguish rejection of a currently correct answer from prediction of failure on an unqueried composition. An initial version was frozen when 266 of the 576 SmolLM3 core responses had been recorded. Input-only validation found that 26 of its 64 true exposure changes were within the accuracy tolerance. Before any challenge inference, a revised version, frozen after 370 core responses, required a true change of at least 0.01. Both versions are archived. This is a dated exploratory extension, not part of the original core protocol; no challenge model outcomes were used in this revision. Neither gate nor its thresholds changed.

For each of the 32 cases, two new compositions add or subtract $0.04d$, where d is a seeded permutation of $(0.75, -0.35, 0.45, -0.85)$ resampled until $|0.04q^\top d| \geq 0.01$. These directions differ from the three audit contrasts. Challenge queries are unavailable to either eight-call gate. A case passes the challenge only if both outputs are valid and within 0.005 absolute exposure error. We report acceptance coverage, accepted cases that fail this subsequent challenge, and rejected cases that pass it. All cases remain paired by their original family; 64 challenge calls per model are not 64 independent cases. This prospective holdout of compositions tests a narrow form of transfer and does not establish broad model reliability.

5.6 Metrics, uncertainty, and controls

Primary exposure accuracy requires an absolute A-weight error at most 0.005, with structurally valid A/B outputs. Risk accuracy uses a 5% relative-error threshold. Invalid and truncated outputs count as failures. We report audit coverage, correct and incorrect accepted answers, and false rejection of a correct base answer; an audit that rejects everything cannot obtain a useful reliability claim by having zero false accepts.

Paired differences use 10,000 bootstrap resamples of the 16 evaluation families, keeping each homogeneous/heterogeneous pair together. The intervals are exploratory and quantify variation in this small designed sample. They are not market-risk confidence intervals. Condition-level calls and shared source documents are never counted as independent observations. We report model-specific results rather than pooling checkpoints into a fictional population of models.

Deterministic controls include exact weighted accounting, classical variance computation, and a grammar-specific parser for the controlled templates. Their success does not establish a need for an LLM. Fault injection, described below, is reported separately from language-model outputs. The release preserves all failures rather than replacing them with calculator answers inside a purported model-response file.

6 Results

6.1 Mechanism validation with injected errors

The executable algebra check enumerates 48 account-weight arrangements and all 80 nonempty vectors in $\{0, 1, 2\}^4$, yielding 3,840 *injected-rule instances*. Of these, 96 have uniform positive multiplicities and preserve effective weights; 3,744 change effective weights. All 3,840 produce exactly the correct homogeneous exposure. For the chosen heterogeneous vector, only the 96 weight-preserving rules produce the exact answer. One contrast detects 3,648 of the 3,744 weighting distortions; the complete three-contrast basis detects all 3,744, with maximum reconstruction error 1.78×10^{-15} in floating-point execution. These are deterministic mechanism checks, not 3,840 model evaluations or estimates of the prevalence of such errors in deployed systems.

6.2 Fresh exposure and risk performance

Main Qwen3 results use the uniform-configuration view described in the execution amendment. All affected responses are replaced according to runtime configuration, never correctness; the original mixed-profile results are reported in Appendix G.

Condition	SmolLM3-3B	Qwen3-8B
Direct aggregate exposure	10/32	13/32
Numerically normalized exposure	13/32	13/32
Split-lot exposure	6/32	6/32
Source-factored exposure	20/32	32/32
Direct risk generation	5/32	2/32
Risk with verified weights	9/32	0/32
Aggregate extraction + calculator	13/32	20/32
Source-factored extraction + calculator	24/32	32/32

Table 1: All-attempt accuracy on 32 cases from 16 paired families per model. Exposure tolerance is 0.005; risk tolerance is 5% relative error. Invalid and truncated outputs count as failures.

For SmolLM3-3B, the paired source-factored minus direct exposure-accuracy difference is 31.2 percentage points (exploratory 95% family-bootstrap interval [9.4, 53.1]). The split-minus-base interval is [-25.0, 0.0] percentage points. The core evaluation has 0 request errors and 0 output-cap terminations in 576 calls.

Among calculated-risk successes for SmolLM3-3B, 3 using aggregate extraction and 4 using source-factored extraction fail the separate exposure-accuracy threshold. Risk and exposure tolerances differ, and risk is not generally an injective function of exposure. These counts demonstrate score disagreement; they do not by themselves identify an exact variance-symmetry mechanism in the observed errors.

For Qwen3-8B, the paired source-factored minus direct exposure-accuracy difference is 59.4 percentage points (exploratory 95% family-bootstrap interval [50.0, 68.8]). The split-minus-base interval is [-37.5, -6.2] percentage points. The core evaluation has 0 request errors and 0 output-cap terminations in 576 calls.

Among calculated-risk successes for Qwen3-8B, 7 using aggregate extraction and 0 using source-factored extraction fail the separate exposure-accuracy threshold. Risk and exposure tolerances differ, and risk is not generally an injective function of exposure. These counts demonstrate score disagreement; they do not by themselves identify an exact variance-symmetry mechanism in the observed errors.

6.3 Audit coverage and incorrect acceptances

Model	Eight-call check	Accepted	Wrong	False reject	Acc. accepted
SmolLM3-3B	Consistency	2/32	0	8	100.0%
SmolLM3-3B	Contrast basis	0/32	0	10	—
Qwen3-8B	Consistency	1/32	0	12	100.0%
Qwen3-8B	Contrast basis	0/32	0	13	—

Table 2: Both checks assess the same base answer. Wrong denotes incorrect accepted base answers; false reject denotes a rejected correct base answer. A dash means no answers were accepted, so conditional accuracy is undefined.

Model	Holdings	Direct	Factored	Audit accepts	Consist. accepts
SmolLM3-3B	identical	9/16	11/16	0	2
SmolLM3-3B	heterogeneous	1/16	9/16	0	0
Qwen3-8B	identical	13/16	16/16	0	1
Qwen3-8B	heterogeneous	0/16	16/16	0	0

Table 3: Stratified descriptive counts. The two holdings types share economic exposure within each family; they are not independent samples.

For SmolLM3-3B, 0/32 cases provide valid outputs on all eight audit queries. Of these, 0 give reconstructed weights outside the admissible simplex tolerance and 0 have base-answer linear-fit error above 0.005. A supplementary source-copy audit finds 18/32 fully faithful ledgers at per-field tolerance 10^{-6}; 2 aggregate-correct ledgers fail that stricter check. This copying check was added during inference and is descriptive, not a frozen primary endpoint.

For Qwen3-8B, 22/32 cases provide valid outputs on all eight audit queries. Of these, 14 give reconstructed weights outside the admissible simplex tolerance and 11 have base-answer linear-fit error above 0.005. A supplementary source-copy audit finds 32/32 fully faithful ledgers at per-field tolerance 10^{-6}; 0 aggregate-correct ledgers fail that stricter check. This copying check was added during inference and is descriptive, not a frozen primary endpoint.

Base tolerance	SmolLM3-3B	Qwen3-8B
0.001	0 / 0	0 / 0
0.005	0 / 0	0 / 0
0.01	0 / 0	0 / 0
0.02	0 / 0	0 / 0

Table 4: Sensitivity analysis: accepted / wrong-accepted counts. The slope threshold is four times the displayed tolerance; curvature and anchor thresholds equal it. Primary correctness labels remain fixed.

Post hoc all-query normalization. The completed SmolLM3 core run has no case with valid raw fractions on all eight audit queries. After inspecting that result, we added a separately dated diagnostic that renormalizes every query output using the same numerical normalization rule as the base-only baseline. This changes values; it is not format-only cleanup. The thresholds remain unchanged and these results do not replace the primary audit.

After normalization, SmolLM3-3B supplies 32 complete diagnostic vectors: 19 reconstruct inadmissible weights and 20 fail the base-answer linear-fit check. These are compatibility diagnostics

Model	Eight valid	Contrast accept/wrong	Consistency accept/wrong
SmolLM3-3B	32	0 / 0	3 / 0
Qwen3-8B	32	0 / 0	1 / 0

Table 5: Explicitly post hoc diagnostic. Correctness is assessed against the normalized base answer. Every existing response is retained, with no threshold tuning or extra model calls.

for the restricted model, not evidence that a particular hidden weighting mechanism has been discovered.

After normalization, Qwen3-8B supplies 32 complete diagnostic vectors: 23 reconstruct inadmissible weights and 19 fail the base-answer linear-fit check. These are compatibility diagnostics for the restricted model, not evidence that a particular hidden weighting mechanism has been discovered.

6.4 Unseen-composition challenge

For SmolLM3-3B, 0/32 cases pass both new composition queries; 10 of the 10 cases with a correct base answer fail at least one challenge query.

For Qwen3-8B, 2/32 cases pass both new composition queries; 11 of the 13 cases with a correct base answer fail at least one challenge query.

Model	Eight-call check	Accept	Fail after	Pass after	Pass but reject
SmolLM3-3B	Consistency	2	2	0	0
SmolLM3-3B	Contrast basis	0	0	0	0
Qwen3-8B	Consistency	1	1	0	2
Qwen3-8B	Contrast basis	0	0	0	2

Table 6: Exploratory transfer test using unchanged core-gate decisions. Fail/pass after means that accepted cases fail/pass the two-query challenge. The queries are withheld from the gate. This is a dated extension designed during core inference, not an independently preregistered benchmark.

6.5 Source-derived holdings results

Model	Direct	Split lots	Source-factored
SmolLM3-3B	0/20	0/20	7/20
Qwen3-8B	0/20	0/20	19/20

Table 7: All three requested exposures must be within 0.0006 of the displayed-data answer. These 20 constructed accounts reuse five factsheets and ten pairs; no independent-source significance claim is made.

Source-tier format failures, complete predictions, and per-target errors are retained in the machine-readable analysis. Correctness concerns ownership of named underlying funds, not security-level diversification or future volatility.

Post hoc unit sensitivity. After both source runs completed, inspection of Qwen3 outputs revealed percentage values in the fractional interface. We therefore separately apply a fixed numerical heuristic: if a target-value block contains a value above one and all entries lie in $[0, 100]$, divide the entire block by 100. For ledgers, apply this per row and separately to the account-weight

vector. Conversion decisions do not consult gold answers; other values remain unchanged. This can misinterpret units in other settings and is not format-only cleanup. It leaves the primary results and tolerance unchanged.

For SmolLM3-3B, this changes 0 direct, 0 split and 1 factored responses. The resulting all-target accuracies are 0/20, 0/20 and 8/20, respectively. These exploratory numbers distinguish visible unit failures from the remaining aggregation errors; they do not establish a generally valid repair algorithm.

For Qwen3-8B, this changes 4 direct, 19 split and 1 factored responses. The resulting all-target accuracies are 0/20, 0/20 and 20/20, respectively. These exploratory numbers distinguish visible unit failures from the remaining aggregation errors; they do not establish a generally valid repair algorithm.

6.6 Query and token cost

The two diagnostic gates use eight calls per case, while source-factored extraction uses one. Table 8 reports the observed resource use; Figure 2 summarizes accuracy on the same controlled cases.

Model	Method	Calls/case	Tokens/case	Sum latency (s)
SmolLM3-3B	Contrast basis	8	2617	83.0
SmolLM3-3B	Consistency	8	2773	83.1
SmolLM3-3B	Source-factored	1	522	66.3
Qwen3-8B	Contrast basis	8	2790	315.4
Qwen3-8B	Consistency	8	3000	325.9
Qwen3-8B	Source-factored	1	519	171.6

Table 8: Recorded prompt plus completion tokens and summed latencies per case. This excludes interrupted attempts without a saved response. Calls run concurrently; summed latency is not wall-clock time. Different model profiles, request batches and host load prevent a controlled timing comparison.

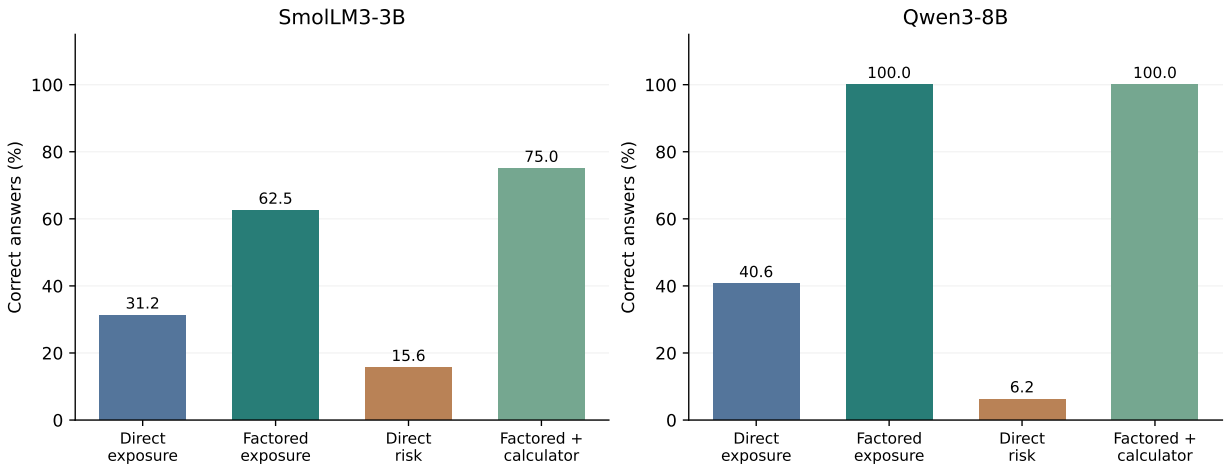


Figure 2: Recorded exposure and risk accuracy. Error bars are omitted because the paired family-bootstrap differences, rather than independent binomial intervals over prompts, are the reported uncertainty analysis.

6.7 Post hoc incompatibility certificates

The full diagnostic and its provenance are specified in Section 4.4. No acceptance threshold was retuned. Table 9 distinguishes strict eligibility from numerical normalization; zero eligible cases do not constitute evidence of compatibility.

Model	View	Eligible	Incompatible	Min. d_*	Median d_*
SmolLM3-3B	strict	0/32	0	—	—
SmolLM3-3B	normalized	32/32	32	0.942	7.060
Qwen3-8B	strict	22/32	22	2.066	7.045
Qwen3-8B	normalized	32/32	32	2.066	8.821

Table 9: Incompatible means that the exact lower bound exceeds 0.005. Minimum and median distances are in percentage points of account exposure; displayed values use certified lower bounds, whose upper-bound gaps are below 3.08×10^{-10} in fractional units.

The SmolLM3 strict view supplies no complete eight-query vector. Qwen3 supplies 22, all incompatible at the reference budget. After numerical normalization, all 32 cases from each model remain incompatible: the smallest required uniform error is about 0.94 percentage points for SmolLM3 and 2.07 for Qwen3. Thus the normalized failure cannot be explained by a single fixed normalized multiplicity vector with answer errors bounded by half a percentage point. This rejects a proposed explanation of the recorded responses, not every possible arithmetic or prompt-dependent explanation.

The component analysis also separates formatting from semantic restrictions. After normalization, all 32 cases per model have valid fractions; none passes the original slope criterion. The curvature criterion passes 0/32 SmolLM3 and 1/32 Qwen3 cases, while the anchor criterion passes 28/32 and 32/32. The zero-coverage result is therefore not solely an interface-validity artifact.

Model	One query	Two queries	Midpoint	Union	Full LP
SmolLM3-3B	23	23	31	32	32
Qwen3-8B	21	25	29	31	32

Table 10: Normalized cases rejected by lower-bound witnesses at the unchanged 0.005 budget, out of 32 per model. The union includes the three displayed families. The LP adds no rejected SmolLM3 case and one Qwen3 case; this does not show general superiority over simple tests.

The release contains 86 exact-checked certificates: 22 strict Qwen3 vectors and 64 normalized vectors, with overlapping observations across views. They are not 86 independent cases. Five new software tests include consistent-but-wrong weights, a contradiction with a known exact optimum, and deliberate certificate corruption; the full suite has 18 passing tests. A separate checker also binds each certificate to archived prompt literals and raw outputs. These are internal checks, not independent human validation.

A global contradiction missed by the three simple witness families. The sole additional normalized Qwen3 case is `fresh_13_identical`. Its base and three positive-contrast responses are shown in Table 11. The base answer is correct. Every one-query, two-query and centered-midpoint witness in the specified families has lower bound zero for this case, yet no nonnegative normalized weighting vector fits all responses exactly.

Query	Product A fractions	Reported A
Base	$(0.296, 0.296, 0.296, 0.296)$	0.296
Contrast 1, plus	$(0.346, 0.346, 0.246, 0.246)$	0.246
Contrast 2, plus	$(0.346, 0.246, 0.346, 0.246)$	0.346
Contrast 3, plus	$(0.346, 0.246, 0.246, 0.346)$	0.346

Table 11: Actual saved Qwen3 responses; their original A/B fractions already sum to one, so normalization leaves these values unchanged. This example is the sole case added by the full LP over the stated simple-witness union.

Write the three positive-contrast input rows as x_1, x_2, x_3. Coordinatewise, $-x_1 + x_2 + x_3 = (0.346, 0.146, 0.346, 0.346)$. Any nonnegative normalized weighting rule must therefore satisfy

$$-f(x_1) + f(x_2) + f(x_3) \leq 0.346. \tag{19}$$

The saved answers instead give $-0.246 + 0.346 + 0.346 = 0.446$. If each answer differs from some fixed rule by at most ϵ, the three discrepancies can account for at most 3ϵ of this gap. Consequently $\epsilon \geq 0.100/3$, or $3.\overline{3}$ percentage points. The LP provides a matching upper bound. This is a falsification of an admissible fixed-rule explanation, not proof that the model literally used negative weights. It strengthens the diagnosis of this finite trace; it does not turn the original zero-coverage gate into a useful correctness verifier.

7 Discussion

For SmolLM3-3B, the contrast audit retains 0 correct base answers and rejects 10 correct ones; the matched consistency check retains 2 correct answers. These counts, together with 0 versus 0 incorrect acceptances, define the observed trade-off. They should not be reduced to a headline about zero false accepts or correctness among a selected subset.

In the unseen-composition extension for SmolLM3-3B, the contrast audit accepts 0 cases, of which 0 later fail the challenge; consistency accepts 2, of which 2 fail. The contrast audit also rejects 0 cases that pass the challenge. These small related samples assess a coverage–transfer trade-off, not a distribution-free reliability guarantee.

For Qwen3-8B, the contrast audit retains 0 correct base answers and rejects 13 correct ones; the matched consistency check retains 1 correct answers. These counts, together with 0 versus 0 incorrect acceptances, define the observed trade-off. They should not be reduced to a headline about zero false accepts or correctness among a selected subset.

In the unseen-composition extension for Qwen3-8B, the contrast audit accepts 0 cases, of which 0 later fail the challenge; consistency accepts 1, of which 1 fail. The contrast audit also rejects 2 cases that pass the challenge. These small related samples assess a coverage–transfer trade-off, not a distribution-free reliability guarantee.

At least one model has zero coverage under the prespecified contrast audit. For that model, the audit is not an empirically useful acceptance policy at this operating point, irrespective of the validity of the conditional identification theorem.

The source-factored baseline separates copying local quantities from multiplying and summing them. Any improvement it shows supports this engineering decomposition on the tested inputs; it does not establish a new general-purpose reasoning algorithm. Conversely, a model that solves the small controlled task limits the scope of claims about widespread fragmentation failures. The complete tables are retained in either case.

The main conceptual distinction is between an answer-level property and an identifiable aggregation rule. Homogeneous fragmentation tests can be useful for detecting some presentation failures, but

passing them cannot establish that each product was weighted correctly. A heterogeneous test adds information, and a full contrast basis identifies the stated fixed-linear error class. That result does not imply that an unrestricted model follows the class.

The operational use of an audit must be assessed through coverage and errors together. An increase in sensitivity obtained by rejecting correct answers is a trade-off, not an unconditional improvement. The matched consistency comparator and the one-call source-factored calculator are therefore essential. A simple, reliable pipeline can be preferable to a more elaborate diagnostic even when the diagnostic has an interesting conditional theorem.

Remaining scope limits. The controlled tier has four products, two underlying assets, three grammars, one response per prompt, and only 16 evaluation families. The source tier contains five related factsheets from one issuer and one date, with PDF layout already removed. Reasoning mode is disabled and the tested models are quantized local checkpoints; this does not characterize current frontier systems, full-precision inference, or tool-using agents. The source tables have internal cross-checks but no independent annotator. No learned model is trained, no trading strategy is tested, and no portfolio return improvement is claimed.

Reproducibility and responsible interpretation. All claims are tied to proof assumptions, injected-rule simulations, or saved inference records. The original pilot is preserved because it documents why the design changed. The release contains the code and numerical source transcriptions needed to inspect the reported results, with model-download manifests rather than redistributed model weights. AI assistance was used in drafting, implementation, and analysis; the named author must review the mathematics, code, sources, and applicable venue disclosure requirements before submission. None of these internal checks substitutes for independent review.

8 Conclusion

Within the original portfolio-fragmentation problem, we identify a specific weakness of apparently successful tests: normalization can make product-weighting errors invisible when product holdings are homogeneous. A contrast basis removes that blind spot for a fixed normalized linear aggregation rule and yields an explicit reconstruction and noise bound. The prespecified contrast gate has zero coverage for both tested models. It therefore does not establish a useful acceptance policy, despite the conditional identification result. The contribution supported by these experiments is a benchmark-design diagnosis and a reproducible assessment of its limits. The post hoc certificate analysis further rules out every fixed convex-combination explanation within the stated error budget for the normalized traces. Its exact certificates strengthen the diagnosis, without supplying a correctness or future-reliability guarantee.

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A Formula provenance and claim ledger

Component	Provenance and defensible claim
Weighted exposure and variance	Standard accounting and classical portfolio mathematics; not invented here.
Multiplicity model	An analytical error class introduced to study this task; not an observation of hidden model computation.
Homogeneous blind spot	A direct algebraic consequence of normalized aggregation, specialized here to a benchmark-design problem. No priority claim for the algebra.
Rank condition and reconstruction	Standard linear identification and Hadamard-basis identities, applied to the stated exposure audit.
Noise bound	A derived operator-norm bound under bounded additive query errors and fixed weights.
Contrast-basis audit	A task-specific candidate diagnostic with recorded model evaluation. It is not a universal certificate or autonomous document verifier.
Source-factored calculator	Established extraction, validation and arithmetic delegation. It is a practical baseline, not a claimed new reasoning paradigm.
Incompatibility certificates	Standard constrained norm approximation and convex duality, specialized to saved exposure responses; exact arithmetic verifies bounds, not model correctness.
Fault-injection counts	Enumerated artificial error rules, never counted as model responses.
Model results	Derived solely from saved responses, with failures and denominators retained.

Table 12: Separating new study contributions from established mathematics and engineering.

B A worked diagnostic example

Let $q = (0.1, 0.2, 0.3, 0.4)$ and suppose the last product is counted twice while the others are counted once. Then $m = (1, 1, 1, 2)$ and

$$p = (1/14, 2/14, 3/14, 8/14).$$

If every product holds 60% A, both $q^\top b$ and $p^\top b$ equal 0.6. The erroneous computation passes the homogeneous exposure test exactly. For the contrast basis in the main text,

$$Cq = (-0.4, -0.2, 0), \qquad Cp = (-4/7, -3/7, 2/7).$$

Their difference is nonzero in every contrast, and Equation 8 recovers $p - q$. This is an analytical example of a specified faulty rule, not an example attributed to an actual model response.

C Why finite probes cannot prove general correctness

Let $\mathcal{Q}$ contain every composition queried by an audit, and let $b^\star \notin \mathcal{Q}$. There exists a function h that is zero at every point of $\mathcal{Q}$ but nonzero at $b^\star$. For example, a sufficiently scaled product of squared distances to the finitely many query points has this property. The function $f(b) = q^\top b + h(b)$ passes all queries in $\mathcal{Q}$ yet disagrees elsewhere. On a compact interior region, scaling can keep the answer within admissible fractional bounds. Therefore the fixed-linear assumption in Proposition 2 is substantive. It cannot be established solely by observing agreement on a finite query set.

Nor do the probes establish the hidden causal mechanism of an LLM. If the model uses different effective weights on different prompts, Equation 8 can return an inadmissible vector or one that

does not reproduce the base answer. We retain those compatibility failures rather than describing every reconstructed vector as the model's actual attention or counting pattern.

A further limitation concerns nonzero acceptance tolerances. Set $q = (0.25, 0.25, 0.25, 0.25)$, $p = (0.2635, 0.2455, 0.2455, 0.2455)$ and $b = (0.9, 0.1, 0.1, 0.1)$. Then $s = (0.018, 0.018, 0.018)$, curvature and anchor error are zero, and all ± 0.05 contrasts remain admissible. Thus the specified fixed-linear rule passes the primary gate, while its base exposure error is $0.0108 > 0.005$. This constructed counterexample is separate from recorded model errors. A smaller slope tolerance could imply a stricter bound under the fixed-linear assumption, but changing the threshold after inspecting evaluation results would require a new prospective study.

D Pilot evidence retained for provenance

The pilot remains a distinct, earlier study: 240 core calls and 32 joint probes for Qwen2.5-1.5B, 120 core calls and 16 joint probes for Qwen2.5-3B, and 32 post hoc requests per model, totaling 472. The initial evaluation contains 16 unique economic cases; the second checkpoint uses eight of them. Numerical inputs include both constructed joint states and observed industry-portfolio returns from the Kenneth French Data Library [7]. The holdings narratives and account allocations are constructed in both tiers.

After a format-only cleanup, all 160 recoverable core risk answers from the 1.5B model equal the illustrative value 1.25. Changing that example to 7.91 produces 7.91 in all eight tested risk outputs. Removing the example still does not give an accurate risk answer in that small post hoc set. Among 16 valid joint reports from the 3B model, 15 report the correct aggregate weights, but none of those 15 gives risk within 5% relative error. These observations concern the emitted reports, not hidden reasoning.

Post hoc numerical renormalization increases exact exposure recovery on the 3B homogeneous eight-product subset from 1/8 to 5/8, while the heterogeneous subset remains 0/8. The four additional successes all map $(0.8, 0.8)$ to a balanced $(0.5, 0.5)$ target. That empirical pattern motivates concern about accidental correctness; it does not demonstrate that Equation 4 describes the model's internal mechanism. The new theorem is a separate analytical characterization.

The full earlier manuscript, all 472 raw responses, prompts, answer keys, and analysis scripts are included unchanged in the archived pilot. Its outcomes are not pooled with the fresh evaluation to estimate a single treatment effect. In particular, new models, cue removal, output-budget changes, and constrained decoding are simultaneous design changes, not a clean one-factor replication.

E Source data and rounding

LifeStrategy fund	World ex-UK	UK all-share	Emerging markets
20%	14.4%	4.0%	1.7%
40%	28.8%	8.0%	3.3%
60%	33.2%	12.2%	5.0%
80%	33.1%	16.2%	6.6%
100%	33.0%	20.0%	8.5%

Table 13: Three target underlying-fund weights, as displayed on page 2 of the five issuer factsheets dated 30 September 2026. Complete disclosed holdings tables and source hashes are in the data package.

The displayed fund weights are not silently normalized. The named targets do not exhaust

the holdings table. Gold account exposure for target a is $q_1 B_{1a} + q_2 B_{2a}$, using the published percentages divided by 100. For example, allocation to an underlying fund means ownership of that fund through the stated wrappers; it does not imply disjoint stock-level holdings among the targets.

Every source record identifies the official URL, PDF page 2, reporting date, and downloaded-byte hash. Only the numerical transcriptions and retrieval metadata are redistributed in the package; the issuer PDFs remain available at their cited sources. The transcriptions are checked by pypdf and Poppler and by in-project visual inspection. This is not independent human annotation and should not be described as such.

F A recorded transfer failure

The following SmolLM3 example is selected by the first case identifier with a correct base answer and a failed challenge, rather than by maximum error. Case `fresh_03_identical` has fixed account weights $q = (0.2, 0.35, 0.05, 0.4)$. The table displays saved numerical answers to four decimal places; full returned text is retained in the release.

Input	Within-product A weights	True A	Reported A	Reported B
Base	(0.6850, 0.6850, 0.6850, 0.6850)	0.6850	0.6850	0.3150
Plus challenge	(0.6510, 0.7030, 0.6710, 0.7150)	0.6958	0.7515	0.2485
Minus challenge	(0.7190, 0.6670, 0.6990, 0.6550)	0.6742	0.7910	0.2810

Table 14: A recorded correct base answer does not transfer to either new composition. The minus response also violates A/B conservation. These are actual model responses, distinct from the specified faulty rule in Figure 1.

G Execution and analysis contract

Model	Records	Slots	Tokens/slot	Repack	RAM cache (MiB)
Qwen3-8B	61	4	1536	off	8192
Qwen3-8B	64	4	2048	on	8192
Qwen3-8B	700	8	1536	off	0
Qwen3-8B	42	8	1536	off	8192
SmolLM3-3B	700	4	4096	on	8192

Table 15: All raw evaluation and execution-recheck records, including the retained initial mixed-profile observations. The main Qwen3 view uses only the final profile. This excludes eight development responses and interrupted unsaved attempts. HTTP payloads specify cache_prompt=false throughout; zero RAM cache also disables the server-level prompt snapshot cache.

The recheck includes every one of the 167 observations recorded outside the final Qwen3 profile. Returned text differs on 41/167 paired requests; this includes formatting changes and is not an error-rate estimate. The 409 original final-profile core responses are reused. The two views therefore are not independent experiments.

The release separates three evidence types: constructed prompts and gold labels, deterministic fault injections, and raw model observations. A raw observation contains the request identifier, timestamp, payload hash, returned text, full response metadata, token usage, finish reason, and

Execution view	Direct	Factored	Direct risk	Audit accepts	Wrong accepts
Initial mixed profile	14/32	32/32	2/32	0	0
Uniform profile	13/32	32/32	2/32	0	0

Table 16: Paired execution sensitivity for the same 32 cases. Both views are retained regardless of which has higher accuracy; the uniform view is used in the main results.

elapsed time. Invalid responses remain in the file. Analysis refuses an incomplete evaluation rather than silently reporting a favorable subset.

The primary fractional-output parser rejects booleans, numeric strings, nonfinite values, negative values, components above one, and nonconservation outside the stated tolerance. The secondary normalization baseline divides nonnegative finite components by their positive sum. It changes numerical values and is explicitly distinguished from format handling. Risk answers must be finite and nonnegative. Truncation is a failure even if a prefix resembles valid JSON.

All new files needed for analysis are under `experiment_v4/`; the unchanged pilot is under `experiment_v3/`. Reproduce numerical analyses without downloading model weights:

```
python -m unittest discover -s experiment_v4 -p 'test_*.py' -v
python experiment_v4/check_baselines.py
python experiment_v4/fault_injection.py
python experiment_v4/analyze.py --model smol
python experiment_v4/analyze.py --model qwen8
python experiment_v4/analyze.py --model qwen8 --uniform
python experiment_v4/analyze_normalized.py --model smol
python experiment_v4/analyze_normalized.py --model qwen8
python experiment_v4/analyze_normalized.py --model qwen8 --uniform
python experiment_v4/analyze_source.py --model smol
python experiment_v4/analyze_source.py --model qwen8
python experiment_v4/analyze_source_units.py --model smol
python experiment_v4/analyze_source_units.py --model qwen8
python experiment_v4/analyze_challenge.py --model smol
python experiment_v4/analyze_challenge.py --model qwen8
python experiment_v4/analyze_challenge.py --model qwen8 --uniform
python experiment_v4/audit_release.py
python build_results.py
python experiment_v4/analyze_certificates.py
python experiment_v4/verify_certificates.py
python build_certificates.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

For new inference, use a separate working copy and archive the supplied result directories first. The runner resumes by request identifier and never overwrites an observed response. Refer to the README for pinned downloads, corpus selection, memory requirements, and recorded runtime variations. The model manifests preserve exact revisions and content hashes; a similarly named model at a different revision is not the same experimental artifact.